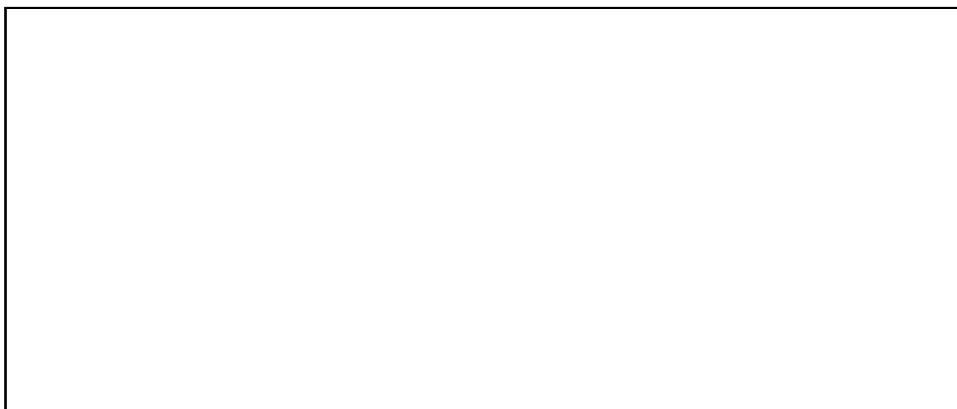


CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
3rd Semester of M.Sc.(Mathematics)Theory Examination November 2018
MA 811:Mathematical Methods-I

Date: 26/11/2018, Monday **Time:** 10:00 a.m. to 10:45 a.m. **Maximum Marks:** 20

Multiple Choice Questions



Instructions:

- (1) Tick the correct answer only in this sheet.
 - (2) You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
 - (3) Each question carries one mark.
 - (4) In this question paper, $\mathbb{R}$ denotes the set of all real numbers and $\mathbb{Z}$ denotes the set of all integers. All notations are standard.
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Q.1. Choose the correct answer for the following questions.

[20]

1. If $F[f(x)] = F(s)$ is the Fourier transform of $f(x)$, then $F[f(x-k)] =$ _____.

- | | |
|-------------------|--------------------|
| a. $e^{-iks}F(s)$ | c. $e^{ks}F(s)$ |
| b. $e^{iks}F(s)$ | d. $e^{-ks}F(s)$. |

2. If $F[f](s)$ is the Fourier transform of $f(x)$ and $f*g$ is the convolution product of functions f and g , then $F[f(x) * g(x)](s) =$ _____.

- | | |
|------------------------------|--------------------------------|
| a. $F[f(x)](s) * F[g(x)](s)$ | c. $F[f(x)](s) + F[g(x)](s)$ |
| b. $F[f(x)](s)F[g(x)](s)$ | d. $F[f(x)](s) - F[g(x)](s)$. |

3. If $F[f(x)](s)$, $F_c[f(x)](s)$ and $F_s[f(x)](s)$ are Fourier transform, Fourier cosine transform and Fourier sine transform of an even integrable function $f(x)$ respectively, then which of the following is true?

- | | |
|----------------------------------|------------------------------------|
| a. $F[f(x)](s) = F_s[f(x)](s)$ | c. $F[f(x)](s) = F_c[f(x)](s)$ |
| b. $F[f(x)](s) = -iF_s[f(x)](s)$ | d. $F[f(x)](s) = -iF_c[f(x)](s)$. |

4. If $F_c[u(x, t)](s)$ is the Fourier cosine transform of $u(x, t)$ in x-variable, then $F_c[\frac{\partial^2 u}{\partial x^2}](s) =$ _____, provided $u(x, t)$ and $\frac{\partial u}{\partial x}(x, t)$ both tend to 0 as $|x| \rightarrow \infty$.

- | | |
|---------------------------|---|
| a. $-s^2 F_c[u(x, t)](s)$ | c. $su(0, t) - s^2 F_c[u(x, t)](s)$ |
| b. $s^2 F_c[u(x, t)](s)$ | d. $-\sqrt{\frac{2}{\pi}}[\frac{\partial u}{\partial x}]_{x=0} - s^2 F_c[u(x, t)](s)$. |

5. If $L^{-1}[\bar{f}(s)]$ is the inverse Laplace transform of $\bar{f}(s)$, then $L^{-1}[e^{-13s}] =$ _____, where $H(t - a)$ is the Heaviside step function and $\delta(t - a)$ is the Dirac delta function.

- | | |
|---------------------|-----------------------|
| a. $\delta(t - 13)$ | c. e^{13t} |
| b. $H(t - 13)$ | d. $\frac{1}{t-13}$. |

6. If $L^{-1}[\bar{f}(s)]$ is the inverse Laplace transform of $\bar{f}(s)$, then

$$L^{-1}\left[\frac{\bar{f}(s)}{s}\right] = \text{_____}.$$

- | |
|-------------------------|
| a. $\frac{f(t)}{t}$ |
| b. $tf(t)$ |
| c. $\frac{d}{dt}(f(t))$ |
| d. $\int_0^t f(k) dk$. |

7. If $L^{-1}[\bar{f}(s)]$ is the inverse Laplace transform of $\bar{f}(s)$, then
 $L^{-1}\left[\frac{1}{s^2 - 64}\right] = \underline{\hspace{2cm}}.$
 a. $\cosh 8t$ b. $\frac{\cosh 8t}{8}$ c. $\frac{\sinh 8t}{8}$ d. $\sinh 8t$.
8. If $L^{-1}[\bar{f}(s)]$ is the inverse Laplace transform of $\bar{f}(s)$, then
 $L^{-1}\left[s^{-\frac{3}{2}}\right] = \underline{\hspace{2cm}}.$
 a. $t^{-\frac{3}{2}}$ b. $t^{\frac{3}{2}}$ c. $\frac{\sqrt{t}}{\Gamma(\frac{1}{2})}$ d. $\frac{\sqrt{t}}{\Gamma(\frac{3}{2})}.$
9. If $L[f(t)]$ is the Laplace transform of a function $f(t)$, then
 $L[H(t - 8)] = \underline{\hspace{2cm}}$, where $H(t - a)$ is the Heaviside step function.
 a. $\frac{e^{8s}}{s}$ b. $\frac{e^{-8s}}{s}$ c. e^{-8s} d. $e^{8s}.$
10. If $f : (0, \infty) \rightarrow \mathbb{C}$ is a periodic function with period μ , then Laplace transform of $f(t)$ is $\underline{\hspace{2cm}}.$
 a. $\frac{1}{1-e^{-s\mu}} \int_0^{\mu} f(t)e^{-st} dt.$ c. $\frac{1}{e^{s\mu}} \int_0^{\mu} f(t)e^{-st} dt.$
 b. $\frac{1}{1-e^{s\mu}} \int_0^{\mu} f(t)e^{-st} dt.$ d. $\frac{1}{e^{-s\mu}} \int_0^{\mu} f(t)e^{-st} dt.$
11. If $L[f(t)]$ is the Laplace transform of a function $f(t)$, then
 $L[e^{5t}t^3] = \underline{\hspace{2cm}}.$
 a. $\frac{3}{(s-5)^4}$ b. $\frac{3}{(s-5)^3}$ c. $\frac{6}{(s-5)^4}$ d. $\frac{6}{(s-5)^3}.$
12. If $L[f(t)]$ is the Laplace transform of a function $f(t)$, then
 $L[\cos 7t] = \underline{\hspace{2cm}}.$
 a. $\frac{7}{s^2+49}$ b. $\frac{s}{s^2+49}$ c. $\frac{7}{s^2-49}$ d. $\frac{s}{s^2-49}.$
13. $\int_{-1}^1 P_{33}(x)P_{37}(x) dx = \underline{\hspace{2cm}}$, where $P_n(x)$ is a Legendre polynomial.
 a. 0 b. $\frac{2}{67}$ c. $\frac{2}{75}$ d. 1.
14. $H_1(50) = \underline{\hspace{2cm}}$, where $H_n(x)$ is a Hermite polynomial.
 a. 0 b. 1 c. 50 d. 100.

15. If $Z[(a_n)](z)$ is the Z-transform of sequence $(a_n)_{n \geq 0}$, then $Z[(1)](z) = \underline{\hspace{2cm}}$.

- a. $\frac{1}{z-1}$ b. $\frac{z}{z-1}$ c. $\frac{1}{z+1}$ d. $\frac{z}{z+1}$.

16. A non empty subset A of an inner product space V with inner product $\langle \cdot, \cdot \rangle$ is called orthogonal if $\underline{\hspace{2cm}}$, for every $x, y \in A$ with $x \neq y$.

- a. $\langle x, y \rangle = 1$ c. $\langle x, y \rangle = 0$
 b. $\langle x, y \rangle = \langle y, x \rangle$ d. none of these.

17. A function $\frac{x^{19} \cos 5x}{x^7 - \sin 3x}$ is $\underline{\hspace{2cm}}$.

- a. an odd function
 b. an even function
 c. neither odd nor even function
 d. a periodic function

18. The period of $\sin \mu t$ is $\underline{\hspace{2cm}}$.

- a. $\frac{2\pi}{\mu}$ c. $\frac{\pi}{\mu}$
 b. 2π d. none of these

19. In Fourier series expansion of $f(x) = x^{101}$ in $[-\pi, \pi]$, $\underline{\hspace{2cm}}$.

- a. only cosine terms are present
 b. only constant term is present
 c. only sine terms are present
 d. both sine and cosine terms are present

20. Consider a 2π periodic function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = x, x \in (-\pi, \pi]$. Then which of the following is true?

- a. Fourier series of f converges to $f(x) = x$ for $x \neq (2n+1)\pi, n \in \mathbb{Z}$
 b. Fourier series of f converges to $f(x) = x$ for $x = (2n+1)\pi, n \in \mathbb{Z}$
 c. Fourier series of f converges to 0 for $x \neq (2n+1)\pi, n \in \mathbb{Z}$
 d. Fourier series of f does not converge.

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3rd Semester of M.Sc.(Mathematics) Theory Examination November 2018
MA 811 Mathematical methods-I

Date: 26/11/2018, Monday **Time:** 10:45 a.m. to 01:00 p.m. **Maximum Marks:** 50

Instructions:

- (5) Section I and II must be written in separate answer sheets
 - (6) You can use only non-programmable calculator. No other gadget is allowed in the examination hall.
 - (7) Figures to the right indicate marks.
 - (8) In this question paper, all notations are standard.
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SECTION-I

Q-2(a) Attempt any TWO of the following: [08]

- (i) Solve $\frac{d^2y}{dx^2} + 9y = x \cos x$ subject to $y(0) = 0, y'(\pi) = 0$ using Green's function.
- (ii) Find the Fourier cosine transform of e^{-kx} . Evaluate the integral
$$\int_0^{\infty} \frac{dx}{(x^2 + 25)(x^2 + 36)}.$$
- (iii) Using inversion integral method, find the inverse Z-transform of $\frac{5z}{(2-z)(3z-1)}.$

(b) Attempt any FOUR of the following: [12]

- (i) Find the half-range Fourier cosine series of $f(x) = x - x^2, 0 < x < \pi.$
- (ii) If the Fourier series of the function $f(x) = |x|$ is $\frac{\pi}{2} - \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\cos(2n-1)x}{(2n-1)^2}$ in $[-\pi, \pi],$ then using Parseval's identity find the sum of the series $\sum_{n=1}^{\infty} \frac{1}{(2n-1)^4}.$
- (iii) If $F[f(x)](s)$ is the Fourier transform of $f(x)$ and $x^n f(x) \in L^1(\mathbb{R}),$ then show that $F[x^n f(x)](s) = i^n \frac{d^n}{ds^n} (F[f(x)](s)).$
- (iv) Define convolution product $f * g$ of functions f and $g \in L^1(\mathbb{R}).$ If $f, g \in L^1(\mathbb{R}),$ then show that $f * g \in L^1(\mathbb{R}).$
- (v) Compute the Fourier series of the function $f(x) = x^2, -\pi \leq x \leq \pi.$

- (vi) Find the Laplace transform of (i) $t \cos 5t$ (ii) $\frac{1-e^t}{t}$.

SECTION II

Q-3(a) **Attempt any THREE of the following**

[18]

- (i) Find the Z-transform of $\{a^n\}_{n \geq 0}$. Solve the difference equation $y_{n+2} - 7y_{n+1} + 12y_n = 0$ subject to $y_0 = 1$ and $y_1 = 2$.
- (ii) Orthonormalize the set $\{1, x, x^2, x^3\}$ in $(C[-1, 1], \langle \cdot, \cdot \rangle)$, where for $x, y \in C[-1, 1]$, $\langle x, y \rangle = \int_{-1}^1 x(t) \overline{y(t)} dt$.
- (iii) Using convolution theorem, find the inverse Laplace transform $\frac{s^2}{(s^2 + 400)(s^2 + 900)}$.
- (iv) Obtain Fourier series for the function $f(x) = \begin{cases} \pi x, & \text{for } 0 \leq x \leq 1, \\ \pi(2-x), & \text{for } 1 \leq x \leq 2. \end{cases}$
Hence show that $f(x) = \frac{8}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)^2} \sin\left(\frac{(2n+1)\pi x}{2}\right)$. Deduce that $\sum_{n=0}^{\infty} \frac{1}{(2n+1)^2} = \frac{\pi^2}{8}$.
- (v) Define Fourier transform of a function $f(x)$. Find the Fourier transform of Gaussian function e^{-kx^2} , $k > 0$.

(b) **Attempt any TWO of the following**

[12]

- (i) Compute the Fourier series of $f(x) = x \sin x$, $-\pi \leq x \leq \pi$. Hence evaluate $\sum_{n=2}^{\infty} \frac{1}{n^2-1}$.
- (ii) Obtain the solution of the free vibration of semi-infinite string governed by the partial differential equations $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$, $0 < x < \infty$, $t > 0$ subject to $u(x, 0) = f(x)$, $\frac{\partial u}{\partial t}(x, 0) = g(x)$ and $u(0, t) = 0$, $t > 0$.
- (iii) Using Laplace transform, solve the differential equation $\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + 5y = e^{-t} \sin t$, where $y(0) = 0$ and $y'(0) = 1$.

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